

Conscious Point Physics: The CPP Apparatus Model: Macroscopic Detectors, Decoherence, and the Quantum Processor Trade-Off Law

Companion SD#3 — Version 1

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Abstract

SD#1 (Abshier, 2026a) established that a measurement apparatus is a CPP structure whose orientation θ couples to the hidden variables λ of the measured particle through the Nexus global constraint. SD#2 (Abshier, 2026b) derived the angular form of this coupling from H_4 symmetry. This paper derives the quantitative aspects: the DP Sea anisotropy imprinted by the apparatus, the decoherence timescales, and the experimental trade-off governing quantum processor tests of CPP.

Four results are established:

Theorem 1 (Classical apparatus): A macroscopic detector with $N_{\text{app}} \sim 10^{26}$ atoms at temperature T decoheres in time $\tau_{\text{dec}} = \hbar/(N_{\text{app}}k_B T) \lesssim 10^{-39}$ s. The apparatus is always classical on any experimentally accessible timescale. This is why quantum mechanics has worked so well for 100 years — not because deviations are impossible, but because macroscopic detectors are never quantum.

Theorem 2 (Quantum processor trade-off): For a quantum processor with N_q qubits at temperature T , the Nexus participation fraction $\varepsilon_{\text{QP}} = N_q/N_{\text{sub}}$ and the quantum coherence time $\tau_q = \hbar/(N_q k_B T)$ satisfy:

$$\varepsilon_{\text{QP}} \cdot \tau_q = \frac{\hbar}{N_{\text{sub}} k_B T} = \text{const}(T).$$

Increasing N_q raises ε_{QP} but proportionally shortens τ_q . The *signal per circuit run* is independent of N_q .

Theorem 3 (DP Sea anisotropy): An oriented apparatus of linear size L imprints a fractional anisotropy $\delta = \text{sea_strength} \cdot (l_P/L) \sim 3 \times 10^{-34}$ on the surrounding DP Sea. This anisotropy is the CPP microscopic description of what “apparatus orientation θ ” means.

Prediction 1 (N-qubit scaling law): For N_q parallel Bell pairs on a quantum processor, the total accumulated CPP signal scales linearly with N_q (because N_q parallel pairs each contribute an independent CPP correction). The functional signature is $\delta E \propto N_q$, while noise grows as $\sqrt{N_q}$. The signal-to-noise ratio for CPP detection scales as $\sqrt{N_q}$, making large-qubit processors the optimal testing ground.

Keywords: apparatus model, decoherence timescale, measurement precision, quantum coherence trade-off, Nexus coupling, Planck-scale decoherence

Plain Language Summary: Quantum measurements require apparatus, and the apparatus itself is made of 600-cell lattice matter. This paper derives a quantitative trade-off between measurement precision and quantum coherence: the more precisely you set a measurement angle, the more the Nexus correlates that setting with the outcome. The decoherence timescale is about 10 to the minus 39 seconds — the apparatus decoheres almost instantly at the Planck scale.

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1 Introduction

SD#1 and SD#2 established the *form* of the CPP deviation from standard Bell correlations. This paper addresses the *scale*: how large is the deviation, why is it undetectable in current experiments, and what experimental architecture gives the best chance of detection?

The answers require a CPP model of the measurement apparatus. In CPP, an apparatus is not an external classical device — it is a collection of $\sim 10^{26}$ eCPs in the DP Sea, participating in the same Nexus constraint as the measured particles. Its orientation θ is a collective degree of freedom of those eCPs. Its thermal decoherence rate determines whether it behaves classically or quantum-mechanically.

The central result is a conservation law (Theorem 2): the product $\varepsilon_{\text{QP}} \cdot \tau_q$ is a constant at fixed temperature. This has the structure of an uncertainty principle for CPP detectability: you can have high Nexus sensitivity (large ε) or long quantum coherence (large τ), but not both. The solution is not more qubits but more *repetitions*: running many short circuits accumulates CPP signal additively while noise only accumulates as $\sqrt{M}$.

2 The Apparatus as a CPP Structure

2.1 Collective orientation in the DP Sea

A polarisation analyser (or spin detector) is, in CPP, a macroscopic collection of N_{app} eCPs arranged in a crystal, magnetic, or optical medium. Its orientation θ is the macroscopic average of the individual eCP SSV vectors:

$$\theta = \frac{1}{N_{\text{app}}} \sum_{i=1}^{N_{\text{app}}} \theta_i, \quad (1)$$

where θ_i is the orientation of the i -th eCP relative to a reference direction. Individual eCPs fluctuate thermally; the collective average is sharp to $O(1/\sqrt{N_{\text{app}}})$.

2.2 DP Sea anisotropy from the apparatus

An oriented collection of eCPs creates an anisotropic SSV field in the surrounding DP Sea. The fractional anisotropy at position $\mathbf{r}$ relative to the apparatus centre is:

$$\frac{\delta\rho_{\text{DP}}(\mathbf{r}, \theta)}{\rho_0} \equiv \delta(\mathbf{r}, \theta) = \text{sea_strength} \cdot \frac{l_P}{|\mathbf{r}|} \cdot \cos(\theta - \phi_{\mathbf{r}}), \quad (2)$$

where $\phi_{\mathbf{r}}$ is the direction angle of $\mathbf{r}$ from the apparatus centre and ρ_0 is the isotropic DP Sea density.

Theorem 2.1 (DP Sea anisotropy scale). *At the surface of the apparatus (characteristic size L), the fractional DP Sea anisotropy is:*

$$\delta_0 = \text{sea_strength} \cdot \frac{l_P}{L} \approx 0.185 \times \frac{1.62 \times 10^{-35} \text{ m}}{L}. \quad (3)$$

For a 1 cm apparatus: $\delta_0 \approx 3 \times 10^{-34}$. This is the CPP microscopic meaning of “the apparatus has orientation θ ”: the DP Sea surrounding the apparatus is anisotropic by the fraction δ_0 in the direction θ .

Proof. The SSV field of a single eCP is proportional to sea_strength times $\hbar c/r^2$ (in natural units where l_P sets the scale). The collective SSV field of N_{app} coherently aligned eCPs scales as $N_{\text{app}} \times l_P/r^2$. At the apparatus surface $r = L$, normalising by ρ_0 and taking the leading dipole term gives (3). $\square$

Remark 2.2 (Why δ_0 is relevant to the Nexus). *The Nexus operates atemporally across the entire lattice. The DP Sea anisotropy δ_0 is the causal imprint of the apparatus on the surrounding DP Sea — the part that propagates via the SSV at c . The Nexus contribution (SD#1 Proposition 1 (Abshier, 2026a)) is additional and atemporal, scaling as $\varepsilon_{\text{Nexus}} = N_{\text{part}}/N_{\text{app}}$. The two contributions are independent channels by which the apparatus orientation influences the particle’s DP Sea state.*

3 Thermal Decoherence of the Apparatus

Theorem 3.1 (Apparatus is always classical). *A macroscopic measurement apparatus with N_{app} atoms at temperature T decoheres to a classical state in time:*

$$\tau_{\text{dec}}^{\text{app}} = \frac{\hbar}{N_{\text{app}} k_B T}. \quad (4)$$

For all experimentally accessible conditions, $\tau_{\text{dec}}^{\text{app}} \ll \tau_{\text{meas}}$, so the apparatus is always in a classical (definite-orientation) state during any Bell measurement.

Proof. The decoherence time (4) follows from the standard Joos–Zeh–Zurek theory of quantum decoherence applied to a macroscopic object (Joos and Zeh, 1985; Zurek, 1991): a superposition of two macroscopically distinct orientations decoheres when the thermal fluctuation energy $N_{\text{app}} k_B T$ exceeds the quantum of uncertainty $\hbar$ divided by the coherence time. The CPP derivation follows by applying the Lindblad master equation of companion 2040d (Abshier and Grok, 2026) to the collective orientation degree of freedom. $\square$

Table 1: Decoherence time (4) for a macroscopic apparatus ($N_{\text{app}} = 10^{26}$ atoms) at various temperatures. All values are far shorter than any measurement timescale ($\tau_{\text{meas}} \gtrsim 10^{-9}$ s).

Condition	T	$\tau_{\text{dec}}^{\text{app}}$
Room temperature	300 K	2.5×10^{-40} s
Liquid nitrogen	77 K	9.9×10^{-40} s
Liquid helium	4.2 K	1.8×10^{-38} s
Dilution refrigerator	10 mK	7.6×10^{-36} s

Remark 3.2 (Why classical detectors do not suppress CPP). *The classical character of the apparatus does not mean the CPP correction vanishes. It means the apparatus orientation θ is sharp and well-defined during the measurement. The Nexus correlation (SD#1, eq. 3) couples θ to the particle hidden variable λ ; this coupling is active regardless of whether the apparatus is in a quantum superposition. The CPP correction exists because the apparatus has a definite orientation, not despite it.*

4 The Quantum Processor: Coherence vs. Participation

4.1 The two roles of qubit count

A quantum processor with N_q qubits at temperature T (dilution refrigerator, $T \sim 10$ mK) plays two roles in CPP Bell tests:

1. **Coherence time:** The qubits form a quantum system with collective decoherence time $\tau_q = \hbar/(N_q k_B T)$, which shrinks as N_q increases.
2. **Participation fraction:** With N_q qubits, the particle-side of the Nexus constraint involves N_q Grid Points, raising the participation fraction to $\varepsilon_{\text{QP}} = N_q/N_{\text{sub}}$, which grows as N_q increases.

These two effects are in direct competition.

Theorem 4.1 (Quantum processor trade-off law). *For a quantum processor with N_q qubits at temperature T , the product of the Nexus participation fraction ε_{QP} and the quantum coherence time τ_q is:*

$$\varepsilon_{\text{QP}} \cdot \tau_q = \frac{N_q}{N_{\text{sub}}} \cdot \frac{\hbar}{N_q k_B T} = \frac{\hbar}{N_{\text{sub}} k_B T} = \text{const}(T). \quad (5)$$

The product is independent of N_q and depends only on temperature and the substrate size N_{sub} .

Proof. Direct substitution of $\varepsilon_{\text{QP}} = N_q/N_{\text{sub}}$ and $\tau_q = \hbar/(N_q k_B T)$. □ □

Remark 4.2 (Uncertainty-principle analogy). *Equation (5) has the structure of an uncertainty principle: increasing N_q raises sensitivity ($\varepsilon_{\text{QP}} \uparrow$) but shortens the coherence window ($\tau_q \downarrow$) by exactly the same factor. There is no optimal N_q in terms of the product; instead, the relevant figure of merit is the total signal accumulated over many circuit runs (see Prediction 4.5).*

4.2 Signal per circuit run

The CPP signal accumulated in a single circuit run of N_{gates} gates is:

$$\mathcal{S}_{\text{run}} = \varepsilon_{\text{QP}} \cdot |f_{H_4}(\theta)| \cdot N_{\text{gates}} = \frac{N_q}{N_{\text{sub}}} \cdot |f_{H_4}| \cdot \frac{\tau_q}{\tau_{\text{gate}}}, \quad (6)$$

where $\tau_{\text{gate}} \approx 50$ ns is the gate time and $N_{\text{gates}} = \tau_q/\tau_{\text{gate}}$. Substituting Theorem 4.1:

$$\mathcal{S}_{\text{run}} = \frac{\hbar}{N_{\text{sub}} k_B T \cdot \tau_{\text{gate}}} \cdot |f_{H_4}(\theta)|. \quad (7)$$

Corollary 4.3 (Signal per run is independent of N_q). *The CPP signal accumulated in a single maximum-depth circuit run is independent of the number of qubits N_q . Numerically, at $T = 10$ mK and $\tau_{\text{gate}} = 50$ ns:*

$$\mathcal{S}_{\text{run}} \approx \frac{1.055 \times 10^{-34}}{10^{26} \times 1.38 \times 10^{-23} \times 10^{-2} \times 5 \times 10^{-8}} \approx 1.5 \times 10^{-28}. \quad (8)$$

Remark 4.4 (What N_q actually buys). *Although N_q does not affect the signal per run, it does help in a different way. With N_q qubits, one can measure N_q parallel Bell pairs simultaneously in*

a single run. Each parallel pair contributes an independent CPP correction. The total signal from N_q parallel pairs per run scales as N_q , while the noise (shot noise) scales as $\sqrt{N_q}$. The signal-to-noise ratio per run therefore scales as $\sqrt{N_q}$ — larger processors are better, but by a square root, not linearly.

4.3 The N_q -scaling law

Prediction 4.5 (Quantum processor signal-to-noise scaling). *For a Bell test on N_q simultaneous qubit pairs, run M times:*

$$\text{Total CPP signal} \sim N_q \cdot M \cdot \mathcal{S}_{\text{run}} \propto N_q M, \quad (9)$$

$$\text{Total noise (shot)} \sim \sqrt{N_q \cdot M}, \quad (10)$$

$$\text{SNR}_{\text{CPP}} \sim \sqrt{N_q \cdot M} \cdot \mathcal{S}_{\text{run}}. \quad (11)$$

A CPP signal, if present, will be identified by its linear growth with N_q at fixed M , distinguishing it from:

- Systematic errors (independent of N_q).
- Shot noise (scales as $\sqrt{N_q}$, not N_q).
- Other quantum effects (would require theoretical identification of why they scale with N_q).

5 Temperature Dependence

From Theorem 4.1, $\varepsilon_{\text{QP}} \cdot \tau_q \propto 1/T$. Colder systems accumulate CPP signal faster. The signal per run (7) scales as $1/T$:

$$\mathcal{S}_{\text{run}}(T) = \frac{\hbar}{N_{\text{sub}} k_B T \cdot \tau_{\text{gate}}} \propto \frac{1}{T}. \quad (12)$$

Table 2: Signal per circuit run (7) vs. temperature, for $N_{\text{sub}} = 10^{26}$ and $\tau_{\text{gate}} = 50$ ns. Cooling from 10 mK to 1 mK gives a factor of 10 enhancement.

Temperature	$\varepsilon_{\text{QP}} \cdot \tau_q$	$\mathcal{S}_{\text{run}}$ (per run, per pair)
100 mK	7.6×10^{-37} s	1.5×10^{-29}
10 mK (current)	7.6×10^{-36} s	1.5×10^{-28}
1 mK	7.6×10^{-35} s	1.5×10^{-27}
0.1 mK	7.6×10^{-34} s	1.5×10^{-26}

Prediction 5.1 (Temperature enhancement). *At fixed N_q and M , the CPP signal scales as $1/T$. Dilution refrigerators at 0.1 mK (achievable but uncommon) give a factor of 100 over current 10 mK systems. Distinguishing CPP from temperature-independent systematic errors is possible by running the same circuit at multiple temperatures: a genuine CPP signal should show the $1/T$ dependence of (12).*

6 The Nexus Coupling $\Gamma(\theta, \lambda)$

Combining the DP Sea anisotropy model (Theorem 2.1) and the Nexus participation fraction (SD#1, Proposition 1), the total coupling of the apparatus orientation θ to the particle hidden variable λ is:

$$\Gamma(\theta, \lambda) = \underbrace{\varepsilon_{\text{Nexus}} \cdot f_{H_4}(\theta - \theta_\lambda)}_{\text{Nexus (atemporal)}} + \underbrace{\delta_0 \cdot \cos(\theta - \theta_\lambda)}_{\text{direct SSV (causal)}}, \quad (13)$$

where θ_λ is the preferred axis of the particle's hidden DP Sea configuration and f_{H_4} is the angular function derived in SD#2.

The direct SSV contribution (second term) falls as $l_P/L \sim 10^{-34}$ and is negligible. The Nexus contribution (first term) at $\varepsilon_{\text{Nexus}} \sim 10^{-26}$ dominates by eight orders of magnitude, confirming that the CPP effect on Bell correlations is dominated by the atemporal Nexus constraint, not by the causal SSV field.

Remark 6.1 (The Nexus term vs. the DP Sea term). *This hierarchy is physically significant. If the CPP correction arose from the direct SSV field (causal), it would propagate at the speed of light and the angular dependence would be $\cos \theta$ — indistinguishable from a systematic error in the standard Bell correlation. Because the Nexus term dominates, the angular dependence is $f_{H_4}(\theta)$ with its 5-fold and 3-fold structure (SD#2). The icosahedral angular signature is the distinctive fingerprint that separates CPP from any causal correction.*

7 Why Standard Bell Tests Are Quantitatively Blind

Combining the magnitude estimate (SD#1) with the apparatus model:

Table 3: Why current Bell tests cannot detect CPP, and what would be needed. $\mathcal{S}_{\text{run}}$ is the signal per circuit run from Corollary 4.3.

Experiment	Sensitivity	CPP signal
Standard optical Bell test	$\sim 10^{-3}$	$\sim 10^{-26}$
Loophole-free Bell (2015)	$\sim 10^{-3}$	$\sim 10^{-26}$
Best precision Bell (2018)	$\sim 10^{-3}$	$\sim 10^{-26}$
Near-future Bell (10^8 runs)	$\sim 10^{-6}$	$\sim 10^{-26}$
10-qubit QP Bell, 10^6 runs	$\sim 10^{-5}$	$\sim 10^{-27}$
1000-qubit QP Bell, 10^9 runs	$\sim 10^{-7}$	$\sim 10^{-25}$
Hypothetical $T = 0.1$ mK system	$\sim 10^{-8}$	$\sim 10^{-26}$

The gap between sensitivity and signal is currently ~ 23 orders of magnitude. No foreseeable technology bridges this gap directly. However, Hossenfelder's prediction (Hossenfelder, 2024) is that quantum computer engineering will produce *unexpected anomalies* — not from directly targeting CPP, but from building ever-larger and more precise quantum systems. The CPP prediction is that those anomalies, when they come, will have the specific angular and scaling signature of Predictions 4.5 and 5.1.

8 Consolidated Open Problems

Open Problem 8.1 (Derive δ_0 from the SSV field equations). *Theorem 2.1 gives the scaling $\delta_0 \sim \text{sea_strength} \cdot l_P/L$ from the SSV dipole field. A rigorous derivation from the CPP SSV field equations (SR/GR companion C2 (Abshier, 2025)) would include the numerical prefactor and the full angular dependence of $\delta(\mathbf{r}, \theta)$ for a realistic polariser geometry (linear vs. circular vs. crystal-face polariser).*

Open Problem 8.2 (Decoherence rate from the CPP Lindblad equation). *Theorem 3.1 quotes the standard Joos–Zeh–Zurek decoherence time. A CPP-specific derivation using the Lindblad master equation of companion 2040d (Abshier and Grok, 2026) would include corrections from the DP Sea structure (the DP Sea has a granular Planck-scale structure that standard quantum field theory ignores). These corrections are $O((l_P/\lambda_{\text{dB}})^2) \sim 10^{-58}$ for room-temperature objects and completely negligible, but the derivation would close the logical gap.*

Open Problem 8.3 (Nexus coupling beyond the participation fraction). *The Nexus participation fraction $\varepsilon_{\text{Nexus}} = N_{\text{part}}/N_{\text{app}}$ is the leading-order estimate of the Nexus coupling strength (SD#1 Proposition 1 (Abshier, 2026a)). A rigorous derivation from the Nexus constraint $\sum_i \delta b_i = 0$ requires integrating the constraint over the apparatus degrees of freedom explicitly. This is the core calculation of SD#4.*

Open Problem 8.4 (Multi-qubit entanglement and the Nexus). *Prediction 4.5 assumes that N_q qubit pairs each contribute independent CPP corrections. If the qubits are mutually entangled (not just pairwise), the hidden variable λ spans all N_q qubits simultaneously. The Nexus coupling between an N_q -qubit entangled state and the N_{sub} -atom substrate may scale differently — potentially as N_q^2/N_{sub} rather than N_q/N_{sub} . Deriving the correct scaling for multi-qubit entangled states is an open problem.*

9 Experimental Programme Summary

Combining SD#1–3, the complete experimental programme for testing CPP is:

Table 4: The three-paper experimental prescription from SD#1–3.

From	What it says	How to test
SD#1	Magnitude $\varepsilon \sim N_{\text{part}}/N_{\text{app}} \sim 10^{-26}$	Baseline; current experiments 22 orders short
SD#2	Angular form: $f_{H_4}(\theta)$ with extrema at $\{36, 72, 120\}$; null at 90	Scan $E(\theta)$ over $[0, 180]$; standard CHSH (45°) is not optimal; use ratio test at H_4 -special angles
SD#3	Signal $\propto N_q$ (parallel pairs), $\propto 1/T$; trade-off law $\varepsilon \cdot \tau = \text{const}$	Use largest available N_q ; cool to lowest T ; run many repetitions; check linear N_q and $1/T$ scalings

The full CPP signature in a quantum processor Bell test is:

1. A residual $\delta E(\theta)$ after subtracting $-\cos \theta$ with 72° and 120° periodicity (SD#2).
2. The residual grows linearly with N_q (SD#3 Prediction 4.5).

3. The residual grows as $1/T$ (SD#3 Prediction 5.1).
4. The ratio $\delta E(36)/\delta E(120) \approx -1.065$ is φ -valued and independent of amplitude (SD#2 eq. 21).
5. The residual vanishes at $\theta = 90$ to leading order (SD#2).

All five properties together would uniquely identify a CPP signal. Properties 2–4 can distinguish CPP from any systematic error with a standard angular dependence.

10 Conclusion

The CPP apparatus model establishes three quantitative results.

The apparatus decoherence theorem (Theorem 3.1) confirms that macroscopic detectors are always classical — consistent with 100 years of Bell tests showing no anomalies. The absence of detected CPP deviations is not a problem for the theory; it is what is expected given the $\sim 10^{-26}$ magnitude of the correction.

The quantum processor trade-off law (Theorem 4.1) shows that the product $\varepsilon_{\text{QP}} \cdot \tau_q = \hbar/(N_{\text{sub}} k_B T)$ is a constant at fixed temperature. More qubits give higher sensitivity but proportionally shorter coherence. The net CPP signal per circuit run is independent of N_q (Corollary 4.3) but the SNR for N_q parallel Bell pairs scales as $\sqrt{N_q}$.

The DP Sea anisotropy (Theorem 2.1) gives the microscopic CPP meaning of apparatus orientation: a fractional $\sim 3 \times 10^{-34}$ anisotropy in the surrounding DP Sea. The Nexus term dominates over the causal SSV term by eight orders of magnitude, so the angular signature is icosahedral (f_{H_4} from SD#2), not cosine-shaped.

The three scaling predictions — linear N_q , $1/T$ temperature dependence, and the $72^\circ/120^\circ$ angular periodicity — together provide a complete, falsifiable experimental signature for CPP quantum foundations.

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